Lecture 19 Scribe

L19.pdf
PDF

Generate a lecture scribe for the given PDF file in chat. This is the lecture file that our professor used while conducting the lecture 19. A lecture scribe is an exam-ready reconstruction of what was taught in the class, and it should allow me to recall definitions and notations, reconstruct proofs and derivations, follow worked examples step by step, and understand logical dependencies. Also, a lecture scribe must not have creative explanations, simplified tutorials, textbook rewrites, a solution manual for unseen problems, and a place to add intuition not discussed in class. The key point is that if it was not in the lecture or textbook context, then it should not appear in the lecture scribe. Also, the purpose of lecture scribing is that it must help me during exams to revise concepts in a shorter period. Hence, for the given lecture pdf, make the lecture scribe that also explains from where the particular formula gets derived.

Here is the **exam-ready Lecture Scribe for Lecture-19** based **only on the given lecture PDF** .
All formulas, logical steps, derivations, definitions, and solved examples are reconstructed exactly in lecture flow.

Lecture Scribe — Lecture 19

N-Variate Joint PDF / PMF / CDF

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1. Joint Distribution of N Random Variables

Definition

For random variables $X_1, X_2, \ldots, X_N$, the vector representation is

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

- The **joint distribution** describes the probability behaviour of all variables **together**.
- It can be represented using:
  - Joint PMF (discrete case)

- Joint PDF (continuous case)
- Joint CDF

(Definition slide — page 17)

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2. Joint PMF

For discrete random variables:

$$P_{X_1,X_2,\ldots,X_N}(x_1,x_2,\ldots,x_N) = P(X_1 = x_1, X_2 = x_2, \ldots, X_N = x_N)$$

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3. Joint CDF

$$F_{X_1,\ldots,X_N}(x_1,\ldots,x_N) = P(X_1 \le x_1, \ldots, X_N \le x_N)$$

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4. Joint PDF

Joint PDF is obtained by differentiating joint CDF:

$$f_{X_1,\ldots,X_N}(x_1,\ldots,x_N) = \frac{\partial^N F_{X_1,\ldots,X_N}(x_1,\ldots,x_N)}{\partial x_1 \partial x_2 \ldots \partial x_N}$$

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5. Independence of Random Variables

Let

$$\mathbf{X} = [X_1, X_2, \ldots, X_N]^T$$

Variables are independent if:

$$f_X(x_1,x_2,\ldots,x_N) = \prod_{i=1}^{N} f_{X_i}(x_i)$$

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Example 1 — Jointly Gaussian Random Variables

Given:

- $\mu_X = 2, \quad \sigma_X^2 = 1$
- $\mu_Y = -3, \quad \sigma_Y^2 = 4$

- $\mathrm{cov}(X, Y) = -1$

Find:

(a) Joint PDF
(b) Marginal PDFs
(c) $P(X < 0)$

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(a) Construction of Joint Gaussian PDF

Step 1: Correlation Coefficient

$$\rho_{XY} = \frac{\mathrm{cov}(X, Y)}{\sigma_X \sigma_Y}$$

From variances:

$$\sigma_X = \sqrt{1} = 1, \quad \sigma_Y = \sqrt{4} = 2$$

Substitute:

$$\rho_{XY} = \frac{-1}{(1)(2)} = -\frac{1}{2}$$

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Step 2: General Joint Gaussian Formula

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho\frac{(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \left(\frac{y -}{c}\right.\right.\right.$$

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Step 3: Substitute Values

$$\mu_X = 2,\ \sigma_X = 1,\ \mu_Y = -3,\ \sigma_Y = 2,\ \rho = -0.5$$

Final Joint PDF:

$$f_{XY}(x, y) = \frac{1}{2\pi\sqrt{3}} \exp\left(-\frac{2}{3}\left[(x-2)^2 + \frac{(x-2)(y+3)}{2} + \frac{(y+3)^2}{4}\right]\right)$$

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(b) Marginal PDFs

Marginals are obtained by integration:

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y)\, dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y)\, dx$$

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Key Property

If variables are jointly Gaussian:

$$X \sim \mathcal{N}(\mu_X, \sigma_X^2), \quad Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$$

Thus:

$$f_X(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x-2)^2}{2}\right)$$

$$f_Y(y) = \frac{1}{2\sqrt{2\pi}} \exp\left(-\frac{(y+3)^2}{8}\right)$$

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(c) Probability $P(X < 0)$

From marginal:

$$X \sim \mathcal{N}(2, 1)$$

Step 1: CDF Relation

$$P(X < 0) = F_X(0)$$

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Step 2: Standardization

$$Z = \frac{X - \mu_X}{\sigma_X}$$

$$P(X < 0) = P\left(Z < \frac{0-2}{1}\right) = P(Z < -2) = \Phi(-2)$$

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Step 3: Symmetry of Normal CDF

$$\Phi(-x) = 1 - \Phi(x) = P(Z > x)$$

Hence:

$$P(Z < -2) = P(Z > 2)$$

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Step 4: Relation to Q-Function

$$Q(x) = P(Z > x)$$

Final Answer:

$$\boxed{P(X < 0) = Q(2)}$$

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Application — Large MIMO System

- System has M transmit antennas and N receive antennas.
- Each channel path coefficient h_{ij} is modeled as a **random variable**.
- Entire system represented by **channel matrix**

$$H = \begin{bmatrix} h_{11} & \cdots & h_{1N} \\ \vdots & \ddots & \vdots \\ h_{M1} & \cdots & h_{MN} \end{bmatrix}$$

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Example 2 — Uniformly Distributed 3D Vector

Region:

$$|x_1| \leq 1, \quad |x_2| \leq 1, \quad |x_3| \leq 1$$

Joint PDF:

$$f(x_1, x_2, x_3) = \begin{cases} C & \text{inside cube} \\ 0 & \text{otherwise} \end{cases}$$

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(a) Find Constant C

Total probability = 1:

$$\int_{-1}^{1}\int_{-1}^{1}\int_{-1}^{1} C\, dx_1 dx_2 dx_3 = 1$$

$$C(2)(2)(2) = 1$$

$$\boxed{C = \frac{1}{8}}$$

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(b) Marginal Joint PDF f_{X_1,X_2}

$$f_{X_1,X_2}(x_1,x_2) = \int_{-1}^{1} \frac{1}{8}\, dx_3 = \frac{1}{4}$$

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(c) Marginal PDF f_{X_1}

$$f_{X_1}(x_1) = \int_{-1}^{1}\int_{-1}^{1} \frac{1}{8}\, dx_2 dx_3 = \frac{1}{2}$$

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(d) Conditional PDF

$$f_{X_1|X_2,X_3} = \frac{f_{X_1,X_2,X_3}}{f_{X_2,X_3}} = \frac{1/8}{1/4} = \frac{1}{2}$$

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(e) Conditional PDF

$$f_{X_1,X_2|X_3} = \frac{1/8}{1/2} = \frac{1}{4}$$

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(f) Independence Check

LHS:

$$f_{X_1,X_2,X_3} = \frac{1}{8}$$

RHS:

$$f_{X_1} f_{X_2} f_{X_3} = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$$

Since LHS = RHS → variables are **independent**.

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Example 3 — Trivariate Exponential-type Distribution

Joint PDF:

$$f_{XYZ}(x, y, z) = \begin{cases} Ke^{-(ax+by+cz)} & x, y, z > 0 \\ 0 & \text{otherwise} \end{cases}$$

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(a) Find K

$$\int_0^\infty \int_0^\infty \int_0^\infty Ke^{-(ax+by+cz)} dx\, dy\, dz = 1$$

Factorization:

$$K \left(\frac{1}{a}\right) \left(\frac{1}{b}\right) \left(\frac{1}{c}\right) = 1$$

$$\boxed{K = abc}$$

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(b) Marginal Joint PDF

$$f_{XY}(x, y) = \int_0^\infty abce^{-(ax+by+cz)} dz$$

$$= abe^{-(ax+by)}$$

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(c) Marginal PDF

$$f_X(x) = \int_0^\infty abe^{-(ax+by)} dy = ae^{-ax}$$

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(d) Independence Check

LHS:

$$f_{XYZ} = abce^{-(ax+by+cz)}$$

RHS:

$$f_X f_Y f_Z = (ae^{-ax})(be^{-by})(ce^{-cz})$$

Thus:

$$\text{LHS} = \text{RHS}$$

Hence X, Y, Z are **independent**.

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✅ End of Lecture-19 Scribe (Exam-Ready)

If you want, I can next:

1️⃣ Make a **super-short revision sheet (2 pages)** from this lecture.
2️⃣ Solve **new exam-level problems from this lecture topic**.
3️⃣ Explain **Joint Gaussian formula derivation more deeply step-by-step (matrix form)**.